

LINE FREQUENCY METER Based On Reciprocal Counting

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Precise measurement of line-signal frequency is very important in many applications, especially in the management of power grid systems. Tasks like calibration of governors of engines that run generators in power plants need a resolution up to 0.01Hz. Frequency is also an important parameter in load sharing among several power plants in the grid.

When the required resolution of the displayed frequency is 1Hz, the instrument becomes simple. Direct frequency counting is one of the many possible methods as shown in Fig. 1.

However, when the resolution required is other than 1Hz, like, say, 0.1Hz or 0.01Hz, value of frequency is obtained from its period. Period is measured in microseconds and fre-

quency is calculated from it. But this method suffers from errors caused by zero-crossing detectors.

In Fig. 2, input conditioning is mainly a zero-crossing detector that produces a pulse for each zero crossing of the sine wave. But here, errors can be caused by comparators whose threshold value may change due to temperature.

Also, hysteresis present in the comparator can cause detection errors. Fig. 3 shows the cause of error in the measurement of the period of a signal.

Reciprocal counter is a class of counter that makes a period measurement on input signal. But it does not use a single- or multi-period counter. Instead, it uses two counters called event and time. Value of frequency can be displayed directly

by taking the reciprocal of the period measured as shown in Fig. 4.

Superiority of the reciprocal counting method over other two methods is given below:

1. Integral cycle of input signal cycles is used so that ± 1 quantisation error is eliminated. This prevents comparator errors from affecting accuracy.

2. Higher resolution is obtained without a long measuring time.

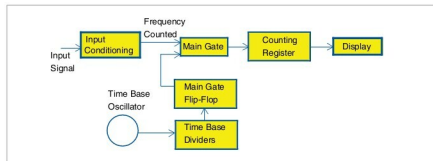


Fig. 1: Block diagram for direct frequency counting

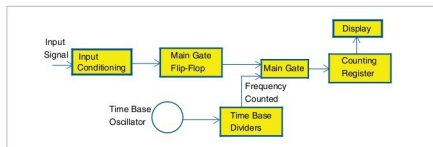


Fig. 2: Block diagram for period measurement

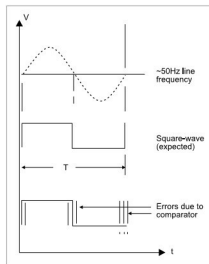


Fig. 3: Zero-crossing errors caused by comparators

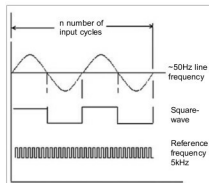


Fig. 4: Diagram showing line-frequency waveform and reference signal